

# OIL INDIA LIMITED - eRTMAC OPERATIONS

OFFICIAL WELL COMPLETION REPORT (WCR) & DRILLING HISTORY  
WELL IDENTIFIER: 16/7-6

## 1. Well Location & Technical Header

|                                               |                                                       |
|-----------------------------------------------|-------------------------------------------------------|
| Operator: Oil India Limited / Joint North Sea | Well Name: 16/7-6                                     |
| UTM Easting (X): 443857.2 m                   | UTM Northing (Y): 6458303.7 m                         |
| Latitude: 58.262225° N                        | Longitude: 2.043161° E                                |
| Total Depth (MD): 2698.4 m                    | Spud Depth: 1110.9 m                                  |
| Drilling Rig: eRTMAC Semi-Submersible Rig-07  | Status: Plugged & Abandoned (P&A;) / Offset Reference |

## 2. Stratigraphic Progression & Formation Tops

| Formation / Stratum Name | Top Depth MD (m) | Lithology Characterization                   | Drillability / Risk Profile |
|--------------------------|------------------|----------------------------------------------|-----------------------------|
| Utsira Fm.               | 1110.9 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Balder Fm.               | 1224.3 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Sele Fm.                 | 1337.7 m         | Shale / Claystone with interbedded siltstone | Tight Hole / Swelling       |
| Lista Fm.                | 1451.1 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Vaale Fm.                | 1564.5 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Ekofisk Fm.              | 1677.9 m         | Chalk / Hard Calcareous Limestone            | Fractured / Losses          |
| Tor Fm.                  | 1791.3 m         | Chalk / Hard Calcareous Limestone            | Fractured / Losses          |
| Hod Fm.                  | 1904.7 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Blodoeks Fm.             | 2018.0 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Roedby Fm.               | 2131.4 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Sola Fm.                 | 2244.8 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Aasgard Fm.              | 2358.2 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Draupne Fm.              | 2471.6 m         | Interbedded Sand / Shale                     | Loss / Depletion            |
| Skagerrak Fm.            | 2585.0 m         | Interbedded Sand / Shale                     | Loss / Depletion            |

## 3. Casing Design & Mud Program Summary

| 36 in Hole     | 30 in Conductor   | 350 m   | Spud Mud / Seawater        | 1.05 SG | FIT 1.25 SG |
|----------------|-------------------|---------|----------------------------|---------|-------------|
| 26 in Hole     | 20 in Surface     | 1,050 m | KCl / Polymer              | 1.12 SG | LOT 1.45 SG |
| 17-1/2 in Hole | 13-3/8 in Interm. | 2,200 m | Low Solids Non-Dispersed   | 1.24 SG | LOT 1.62 SG |
| 12-1/4 in Hole | 9-5/8 in Prod.    | 2398 m  | Oil Based / Synthetic Mud  | 1.32 SG | LOT 1.78 SG |
| 8-1/2 in Hole  | 7 in Liner        | 2698 m  | Synthetic Oil Based (SOBM) | 1.38 SG | LOT 1.88 SG |

4. Historical Operational Incidents, NPT & Mitigations (Well: 16/7-6)

*This section constitutes the institutional memory log extracted by the NWIS knowledge engine. All alerts generated during active drilling cite these exact records.*

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------|
| <b>INCIDENT #1: Lost Circulation / Mud Loss</b>                                                                                                                                                        |                            |                                 |
| <b>Severity:</b> HIGH                                                                                                                                                                                  | <b>NPT Hours:</b> 21.0 hrs |                                 |
| <b>Depth Interval:</b> 2480.0 m - 2540.0 m                                                                                                                                                             | <b>Formation:</b> CHALK GP | <b>Status:</b> Resolved on Site |
| <b>Event Occurrence Narrative:</b><br>Lost 450 bbls synthetic mud at 2510m in fractured Ekofisk chalk. Static loss rate 40 bbl/hr with zero returns on dynamic circulation.                            |                            |                                 |
| <b>Mitigation Action Applied:</b><br>Bullheaded 60 bbls dual-particle size calcium carbonate LCM pill (D50=450um). Reduced mud weight from 1.28 SG to 1.21 SG. Successfully regained full circulation. |                            |                                 |
| <b>Offset Drilling Recommendation &amp; Institutional Lesson:</b><br>Ekofisk chalk fractures easily if surge pressures exceed 0.04 SG equivalent. Control tripping speeds to max 15 m/min.             |                            |                                 |

|                                                                                                                                                                                                                |                            |                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------|
| <b>INCIDENT #2: Cementing Channeling / Loss</b>                                                                                                                                                                |                            |                                 |
| <b>Severity:</b> MEDIUM                                                                                                                                                                                        | <b>NPT Hours:</b> 16.0 hrs |                                 |
| <b>Depth Interval:</b> 2620.0 m - 2680.0 m                                                                                                                                                                     | <b>Formation:</b> TOR FM   | <b>Status:</b> Resolved on Site |
| <b>Event Occurrence Narrative:</b><br>Loss of returns during 9-5/8 inch casing cementing job after pumping 120 bbls lead slurry. Annular pressure drop indicated loss into Tor formation vugs.                 |                            |                                 |
| <b>Mitigation Action Applied:</b><br>Dropped top plug and displaced with rig pumps at reduced rate. Top of cement found 180m below target. Performed remedial squeeze cementing through perforations at 2615m. |                            |                                 |
| <b>Offset Drilling Recommendation &amp; Institutional Lesson:</b><br>Use thixotropic lightweight lead cement slurry (1.45 SG) across Tor chalk formation to avoid exceeding fracture gradient.                 |                            |                                 |